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Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Code: 104604 | ***Subject: Electronics Instruments and Measurement*** | ***Time: 03 Hours*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): **[14]**

(a) The high torque to weight ratio in an analog indicating instrument indicates
- (i) High Friction loss
- (ii) Low friction loss
- (iii) Nothing as regards friction
- (iv) None of the above

(b) Which of the following is the most popular method for measuring low resistance?
- (i) ducter ohmmeter method
- (ii) kelvin double bridge method
- (iii) ammeter-voltmeter method
- (iv) potentiometer method

(c) Which of the following method is used for the measurement of Medium Resistance?
- (i) Direct-Deflection method
- (ii) Anderson Bridge
- (iii) Carey-Foster bridge method
- (iv) Kelvin's double bridge method

(d) Which of the following is a balance equation for computing the resistance?
- (i) $R_1 = R_2/R_4$
- (ii) $R_1 = R_3/R_4$
- (iii) $R_1 = R_2R_3$
- (iv) $R_1 = R_2R_3/R_4$

(e) A Schering bridge can be used for
- (i) measuring voltages
- (ii) testing capacitors
- (iii) protecting the circuit from temperature rises
- (iv) measuring currents

(f) What is the dependence of frequency on the balance equation?
- (i) varies by a factor of 2
- (ii) depends on the detector used
- (iii) independent
- (iv) depends on the supply magnitude

(g) Which of the following can be measured using Maxwell's Inductance Bridge?
- (i) capacitance
- (ii) frequency
- (iii) mutual Induction
- (iv) Induction

(h) The windings of a C.T. are
- (i) tied together
- (ii) Shorted
- (iii) wound over one another
- (iv) Grounded

(i) Which of the following device is used to measure power in A.C. circuits?
- (i) ammeter
- (ii) Wattmeter
- (iii) voltmeter
- (iv) ohmmeter

(j) When the moving coil in a Dynamometer type wattmeter deflects
- (i) pointer doesn't move
- (ii) current flows
- (iii) voltage is generated
- (iv) pointer moves

Q.2 Solve both questions : **[14]**

(a) Explain how the unknown frequency is measured using Wien bridge method?

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(b) What is the need for standards of measurements? How are they classified?

Q.3 Solve both questions : **[14]**

(a) How do you measure the unknown inductance using Hay's Bridge?

(b) State the purpose of calibration of measuring instruments. Explain the procedure of calibration of D.C. Voltmeter by using D.C. Potentiometer.

Q.4 Solve both questions : **[14]**

(a) How do you measure the unknown capacitance using Schering Bridge? Also draw the phasor diagram.

(b) Draw the possible methods of connecting the pressure coil of a wattmeter and compare the errors. Explain the meaning of 'compensating winding' in a wattmeter and show how they help to reduce the error.

Q.5 Solve both questions : **[14]**

(a) List out comparisons between CT's and PT's (Any four).

(b) A current transformer with a bar primary has 300 turns in its secondary. Resistance and reactance of the secondary circuit are 1.5 Ω and 1.0 Ω respectively. Including the transformer winding with 5 A flowing in the secondary winding, find the phase angle error. (Assuming magnetizing mmf to be 100 AT and core loss to be 1.5 W).

Q.6 Solve both questions : **[14]**

(a) Describe the construction and principle of operation of D' Arsonval type Galvanometer.

(b) Write short notes on (a) Kelvin's double bridge (b) Digital frequency meter.

Q.7 Solve both questions : **[14]**

(a) Describe the measurement of frequency, phase angle and time delay using oscilloscope with suitable diagrams and mathematical expressions.

(b) A moving coil instrument gives a full scale deflection of 10mA, when the potential difference across its terminal is 100mV. Calculate (1) The shunt resistance for a full scale deflection corresponding to 100A (2) The resistance for full scale reading with 1000V. Calculate the power dissipation in each case?

Q.8 Solve both questions : **[14]**

(a) Draw the block diagram of an electronic voltmeter and explain its operation.

(b) Explain BH curve with a neat diagram. What is Hysteresis loop?

Q.9 Solve both questions : **[14]**

(a) Prove that for electrodynamometer type wattmeter true power = $[\cos\Phi] / [\cos\Phi\cos(\Phi-\beta)]\times$*actual wattmeter reading*. Where cosφ = power factor of the circuit, $\beta=\tan^{-1}(wL/R)$ where L and R are the inductance and resistance of the pressure coil.

(b) Explain with a neat sketch, the construction and working principle of Megger.

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